

AI Exposure and Wage Growth in Australia: 2020–2024

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GitHub repository link:

<https://github.com/TingyingHuang/ECC3479-The-impact-of-AI-exposure-on-wage-growth.git>

Abstract

This paper documents the conditional association between occupation-level AI exposure and wage growth in Australia over 2020–2024. Using a two-way fixed effects event-study model estimated on 7,912 individuals from the HILDA Survey, I find that workers in higher-AI-exposure occupations in 2020 are associated with lower within-person wage growth in every subsequent year. By 2023, a one-standard-deviation higher AI exposure score corresponds to approximately 4.8 percentage points lower within-person wage growth, conditional on individual fixed effects and common time trends. The association is robust across standard error formulations, control variable sets, sample restrictions, and functional forms. The analysis is descriptive: the 2020 reference year coincided with anomalous pandemic-driven demand for AI-adjacent workers, so the estimated compression cannot be attributed to AI dynamics alone. An alternative base year check confirms the compression persisted beyond the immediate post-pandemic adjustment.

1 Introduction

How is occupational AI exposure associated with wage growth across all occupations in Australia over 2020–2024, and how has that association changed year by year? Two considerations motivate the question. First, AI systems can either substitute for or complement the tasks concentrated in high-exposure occupations, and the two channels have opposite implications for relative wages (Autor, 2015). Substitution displaces labour demand and compresses wages in exposed occupations; complementarity raises productivity and could lift them. Which effect dominates in practice, and by how much, is the central empirical question for any assessment of AI’s labour market impact. Second, while most relevant research currently focuses on the United States, Australia’s distinct labour market institutions make a direct local estimate independently valuable.

To answer this, I match HILDA survey data with AI Occupational Exposure (AIOE) scores from Felten et al. (2021), mapped to Australian occupations via occupation crosswalks, and construct a longitudinal panel of 7,912 workers. Fixing each worker’s AI exposure to their 2020 occupation avoids endogenous sorting confounds; a TWFE event-study model with individual and year fixed effects then traces the year-by-year association. The results show that workers in higher-AI occupations fell progressively behind: by 2023, a one-standard-deviation higher AI exposure score is associated with -0.206 log points lower wage growth ($SE = 0.030$, $p < 0.001$), equivalent to roughly 4.8 percentage points, before stabilising in 2024. This pattern holds across alternative standard errors, control sets, sample restrictions, and functional forms.

2 Data

2.1 Sources of data

Individual-level data come from the Household, Income and Labour Dynamics in Australia (HILDA) Survey, an annual nationally representative longitudinal household survey conducted by the Melbourne Institute since 2001 (Melbourne Institute, 2024). The survey records annual wages, two-digit ANZSCO occupation codes, highest education qualification, age, sex, and state of residence.

AI exposure scores are taken from Felten et al. (2021), who construct an AIOE index by rating the relevance of 52 AI application areas to occupational task requirements using O*NET data; the output is a continuous score per US SOC-2010 occupation code. To apply these scores to Australian workers, I map them to ANZSCO 2-digit codes via a two-step crosswalk: SOC-2010 $\rightarrow$ ISCO-08 (ILO, 2012) $\rightarrow$ ANZSCO 2-digit (ABS, 2022). The crosswalk covers 99.6% of valid occupation-year observations.

2.2 Sample construction

The starting point is all HILDA respondents with at least one observation between 2020 and 2024 (14,399 persons). Four sequential restrictions then narrow the sample to an estimation-ready panel. (i) Observation-years with missing log wage, AI exposure score, or education are dropped (14,296 persons remain); this step also removes the $\approx 0.4\%$ of respondents whose 2020 ANZSCO code is unmatched in the crosswalk and therefore carries no AI score. (ii) Respondents without a valid 2020 observation are next dropped, since the baseline AI exposure must be anchored to that year (8,864 persons remain); these are largely workers who were not employed or surveyed in 2020. (iii) Respondents with fewer than two years of valid observations are then excluded, as individual fixed effects require within-person variation (7,913 persons remain). (iv) One further respondent is dropped due to an unmappable education code. The final estimation panel contains 7,912 individuals and 32,664 person-year observations over 2020–2024 (24,752 in the 2021–2024 outcome window; 2020 serves as the reference year).

2.3 Key variables

The outcome is log annual wages ($\log w_{it}$). Using logs is conventional in wage equation literature and has a practical advantage here: it normalises the right-skewed wage distribution and lets coefficients be read directly as approximate percentage changes. As the treatment variable, I use each worker’s AIOE score fixed to their 2020 occupation (AI_i), a continuous index ranging from -0.53 to 0.52 with a sample mean of 0.086 and standard deviation of 0.234. Fixing exposure to 2020 avoids endogenous sorting: workers who move to higher-paying occupations after AI adoption would otherwise bias the estimated effect upward. One control variable is included: an ordinal education measure coded 1 (postgraduate degree) to 7 (Year 11 or below), where higher values indicate lower qualifications. Education enters the model because 9.1% of individuals change their highest qualification during the panel, and it correlates with both AI exposure and wages.

Table 1: Table 1: Summary Statistics — Main Analysis Sample (baseline-fixed, 2020–2024)

Variable	Mean	SD	Min	Median	Max	N
Log annual wage	7.175	0.715	0.693	7.244	11.000	32,664
AI exposure (2020 baseline)	0.086	0.234	-0.531	0.097	0.517	32,664
Education (ordinal 1–7)	3.775	1.798	1.000	3.000	7.000	32,664
Age	41.3	13.6	15.0	40.0	82.0	32,664

Note: Statistics computed over the full 2020–2024 panel. SD = standard deviation; N = number of non-missing person-year observations. Education: 1 = postgraduate degree; 7 = Year 11 or below.

Table 1: produced by `output/primary_analysis.ipynb`.

3 Empirical Strategy

The primary specification is a two-way fixed effects (TWFE) event-study model with a continuous treatment (Angrist & Pischke, 2009):

$$\log w_{it} = \sum_{t \in \mathcal{T}} \beta_t \cdot (\text{AI}_i \times \mathbf{1}\{\text{year} = t\}) + \gamma \cdot \text{edu}_{it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where $\mathcal{T} = \{2021, 2022, 2023, 2024\}$, AI_i is the 2020 baseline AIOE score, μ_i are individual fixed effects, λ_t are year fixed effects, and standard errors are clustered at the individual level. Each $\hat{\beta}_t$ measures the conditional association between AI exposure and within-person log wages in year t relative to the 2020 reference year.

3.1 Descriptive analysis

This analysis estimates conditional associations. Because confounding shocks occurring within the same period cannot be fully ruled out, the results are descriptive rather than causal. That said, the model controls for two broad classes of confounders. The individual fixed effects μ_i absorb all time-invariant individual characteristics that jointly determine AI exposure and wages: ability, career orientation, stable occupational sorting, and demographic factors. The year fixed effects λ_t absorb aggregate shocks affecting all workers uniformly, including inflation, the business cycle, and the post-COVID normalisation. Identification comes from the differential within-person wage trajectory across workers at different positions in the 2020 AI exposure distribution.

3.2 The assumption required and the evidence for it

For the results to be meaningful, the wage advantage of high-AI occupations over low-AI ones should not have been trending before 2020. I estimate the cross-sectional slope of wages on AI exposure year by year from 2001 to 2019: it holds steady at around 0.59 (range 0.49–0.69) with no sign of acceleration, so the post-2020 compression is not a continuation of something already in motion.

AI exposure is fixed to each worker’s 2020 occupation rather than updated annually because 37.5% of workers change major occupation group over the panel. Letting the score update would mix the wage effect of AI with the wages workers earn after switching jobs; the sign reversal in the time-varying specification (+0.084 vs. −0.091 in 2021) shows how large that distortion would be.

4 Results

Table 2 reports the main TWFE estimates, with individual and year fixed effects, an education control, and standard errors clustered at the individual level.

Table 2: TWFE Event-Study: AI Exposure and Log Wages, 2021–2024

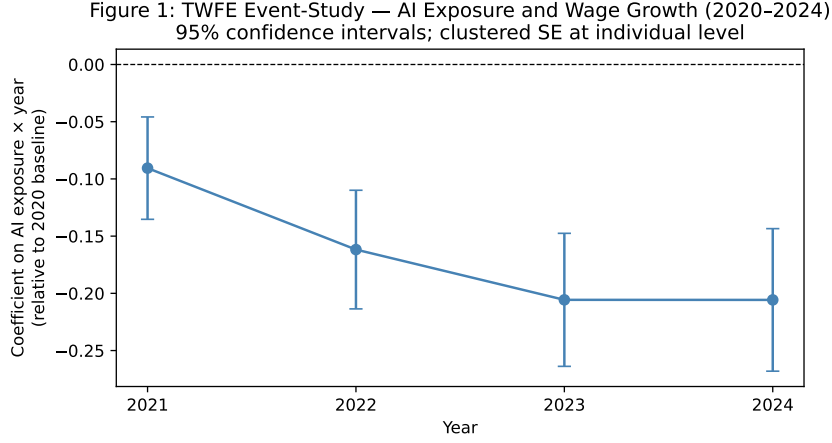
	Coefficient	SE
AI exposure $\times$ 2021	-0.0906***	(0.0228)
AI exposure $\times$ 2022	-0.1617***	(0.0264)
AI exposure $\times$ 2023	-0.2057***	(0.0297)
AI exposure $\times$ 2024	-0.2057***	(0.0318)
Education (ordinal)	0.2205***	(0.0131)
Individual FE	Yes	
Year FE	Yes	
SE type	Clustered (individual)	
N (obs)	32,664	
Individuals	7,912	
Within- R^2	0.159	

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Dependent variable is log annual wage; reference year is 2020. AI exposure is fixed to each worker's 2020 occupation.

Table 2: produced by `output/primary_analysis.ipynb`.

4.1 Direction and magnitude

Workers in higher-AI occupations saw lower wage growth in every year after 2020. Controlling for individual fixed effects, year fixed effects, and education, a one-standard-deviation higher AI exposure score is associated with approximately 4.8% lower within-person wage growth by 2023 ($p < 0.001$). To put that in concrete terms: the gap between a laboratory technician and a software developer in AI exposure is roughly one standard deviation, and by 2023 that gap translated into a 4.8 percentage-point difference in wage growth. The raw interaction coefficients are -0.091 (2021), -0.162 (2022), -0.206 (2023), and -0.206 (2024) — the penalty grew each year through 2023, then held steady in 2024.



Source: output/primary_analysis.ipynb.

Figure 1: X-axis: survey year (2021–2024). Y-axis: TWFE coefficient on AI exposure $\times$ year, relative to 2020 baseline (log wage units). Error bars are 95% confidence intervals.

4.2 Economic significance

The average year-on-year within-person log wage change over 2021–2024 is approximately 8.5% per year. A 4.8 percentage-point gap represents roughly 0.6 times that average: high-AI workers are falling behind by more than half of the typical annual wage increment, conditional on their own prior trajectory and the economy-wide wage trend. The gap widened monotonically from 2021 (−0.091) to 2023 (−0.206) and then held steady in 2024 (−0.206), suggesting a persistent divergence rather than a one-off adjustment.

5 Robustness Checks

Table 3 presents nine specifications, each varying one element from the baseline.

The time-varying specification is the most informative: the 2021 coefficient flips from −0.091 to +0.084 when AI exposure is updated annually, reflecting endogenous job-switching rather than a reversal of the AI wage effect; from 2022 onward, both specifications agree on direction.

Everything else passed. The pre-trend check shows the cross-sectional AI–wage slope was stable from 2001 to 2019 with no sign of drift, supporting the parallel trends assumption. Alternative standard errors, control sets, sample restrictions, functional form, alternative base year, and an age placebo all return negative and significant 2022–2024 coefficients or, in the placebo case, zero.

Table 3: Robustness Summary: AI $\times$ Year Coefficients across Nine Specifications

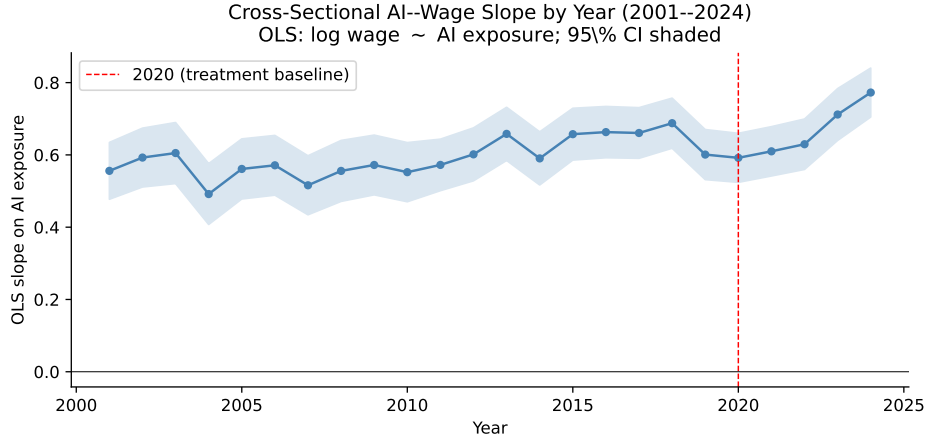
	(1) Main	(2) Time-vary	(3) HC3 SE	(4) No edu	(5) +State	(6) Trim 1%	(7) Non-sw.	(8) Prime-age	(9) IHS wage
AI $\times$ 2021	-0.091*** (0.023)	0.084*** (0.023)	-0.091*** (0.021)	-0.097*** (0.023)	-0.090*** (0.023)	-0.036 (0.020)	-0.033 (0.025)	-0.026 (0.027)	-0.091*** (0.023)
AI $\times$ 2022	-0.162*** (0.026)	0.003 (0.024)	-0.162*** (0.022)	-0.174*** (0.027)	-0.160*** (0.026)	-0.112*** (0.023)	-0.082** (0.029)	-0.052 (0.030)	-0.162*** (0.026)
AI $\times$ 2023	-0.206*** (0.030)	-0.008 (0.027)	-0.206*** (0.024)	-0.226*** (0.031)	-0.204*** (0.030)	-0.151*** (0.025)	-0.090** (0.031)	-0.073* (0.031)	-0.206*** (0.030)
AI $\times$ 2024	-0.206*** (0.032)	-0.041 (0.028)	-0.206*** (0.025)	-0.231*** (0.033)	-0.205*** (0.032)	-0.161*** (0.028)	-0.095** (0.035)	-0.043 (0.032)	-0.206*** (0.032)
Indiv. FE	Yes								
Year FE	Yes								
SE type	Clust.	Clust.	HC3	Clust.	Clust.	Clust.	Clust.	Clust.	Clust.
Controls	Preferred	Preferred	Preferred	AI only	Edu+State	Preferred	Preferred	Preferred	Preferred
Outcome	$\log w$	$\log w$	$\log w$	$\log w$	$\log w$	$\log w$	$\log w$	$\log w$	$\sinh^{-1} w$
AI measure	2020 fixed	Time-vary	2020 fixed	2020 fixed	2020 fixed	2020 fixed	2020 fixed	2020 fixed	2020 fixed
Observations	24,752	33,244	24,752	24,752	24,752	24,317	14,393	17,479	24,752
Individuals	7,912	10,881	7,912	7,912	7,912	7,860	4,738	5,967	7,912
Within- R^2	0.159	0.158	0.159	0.121	0.160	0.159	0.107	0.127	0.159

Note: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Standard errors in parentheses; clustered at individual level except Column (3). Each column changes exactly one element from Column (1). Column (2) uses time-varying rather than 2020-fixed AI exposure; the 2021 sign reversal is expected under endogenous sorting. Column (9): outcome is $\sinh^{-1}$ (annual wage); coefficients are not directly comparable in magnitude to Columns (1)–(8).

Table 3: produced by `output/robustness_checks.ipynb`.

5.1 Pre-trend check, 2001–2019

The TWFE parallel trends assumption requires that, absent any differential AI-related shock, high- and low-AI workers would have followed similar wage trajectories. I test this by estimating the cross-sectional OLS slope of log wages on AI exposure year by year from 2001 to 2019: if that slope was already trending before 2020, the post-2020 compression could be a continuation of a pre-existing pattern rather than a structural break. The slope sits around 0.59 every year with no sign of drift before 2020. Level differences between high- and low-AI occupations may exist, but the TWFE design identifies from changes relative to each worker’s own trajectory, not levels.

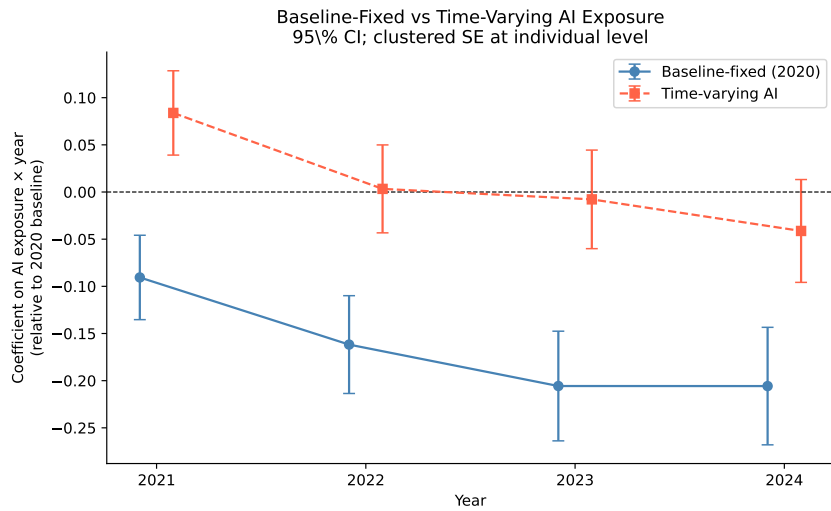


Source: `output/robustness_checks.ipynb`.

Figure 2: Cross-sectional OLS slope of log wage on AI exposure, estimated separately for each year 2001–2024. Shaded band is 95% confidence interval.

5.2 Time-varying AI exposure

Fixing AI exposure to 2020 is a deliberate design choice to avoid endogenous sorting bias: if workers move to higher-paying occupations after AI adoption, a time-varying score would conflate the wage effect of AI with the wages that job-changers select into. The time-varying specification relaxes this, updating each worker’s AI score annually, and serves as a direct test of how sensitive the results are to this identification choice. When AI exposure is updated annually, 2021 flips from -0.091 to $+0.084$ — workers who moved into high-AI roles in 2021 did so partly for higher pay, exactly the sorting bias the baseline design aims to avoid. From 2022 onward, both specifications agree on sign, though the time-varying estimates are smaller, consistent with residual sorting confounding the annual scores.



Source: `output/robustness_checks.ipynb`.

Figure 3: Baseline-fixed (circles) vs. time-varying (squares) AI exposure coefficients. X-axis: year. Y-axis: TWFE coefficient on AI exposure $\times$ year. Error bars are 95% confidence intervals.

5.3 Additional specification checks

Seven further checks each vary one element of the baseline specification; all return negative and significant 2022–2024 coefficients. A separate placebo check uses age as the outcome and returns essentially zero.

Standard errors (HC3). Tests whether significance depends on the clustering assumption. All p -values remain below 0.001; inference is not an artefact of clustering.

No education control. Education correlates with both AI exposure and wages, so omitting it tests the direction of omitted-variable bias. The 2023 coefficient shifts from -0.206 to -0.226 ; higher-AI workers tend to have lower qualifications, so the AI term absorbs part of that penalty when education is excluded. Direction and significance are unaffected.

State dummies. High-AI occupations cluster geographically, so state-level demand shocks could confound the estimate. Adding eight state fixed effects shifts the 2023 coefficient by less than 0.002 log points.

Wage trimming. Tests sensitivity to outlier wages that could have outsized leverage in OLS. Removing the top and bottom 1% of log wages leaves the 2023 coefficient stable.

Non-switchers. Occupation changers could introduce reverse causality; restricting to the 4,738 workers who never changed major occupation group removes that channel. The 2023 coefficient is -0.090 ($p < 0.01$) — smaller but still negative and significant, so the result is not purely compositional.

Prime-age workers (25–55). Life-cycle wage dynamics at the age extremes could correlate with AI exposure. Restricting to prime-age workers gives -0.073 in 2023 ($p < 0.05$) — attenuated but signed the same way.

Functional form (IHS). Tests whether results depend on the log transformation. Using $\sinh^{-1}(\text{raw wage})$ yields a 2023 coefficient of -0.206 , matching the sign and pattern of the log specification.

Alternative base year (2021). Since 2020 was anomalous, re-anchoring to 2021 tests whether compression persisted beyond the pandemic adjustment. The 2023 coefficient is -0.133^{***} ($\text{SE} = 0.026$); the result holds.

Placebo (age as outcome). Tests whether the TWFE mechanics generate spurious significance regardless of the outcome. Using age returns $\approx 10^{-14}$, ruling this out.

6 Discussion, Limitations, and Conclusion

6.1 What this analysis can and cannot conclude

Workers in higher-AI occupations fell behind in wage growth every year from 2021 to 2024, with the gap reaching 4.8 percentage points per standard deviation of AI exposure by 2023 — about 0.6 times the average annual increment. This held across all robustness checks and among occupation non-switchers (-0.090 in 2023), so it is not a story about who changes jobs. High-AI occupations in Australia have, by any reasonable read of the estimates, fallen behind in wage growth since 2020.

What the analysis cannot establish is why. Individual fixed effects absorb stable worker characteristics and year fixed effects absorb aggregate trends, but within-period confounders remain. The 2020 reference year is the main problem: pandemic demand inflated wages in AI-adjacent occupations in exactly the year used as the baseline, so some of the compression likely reflects that premium reversing. Separately, high-AI occupations overlap with the technology sector, which faced independent demand corrections in 2022 — shocks that individual fixed effects do not absorb.

6.2 Limitations and what would resolve them

Re-anchoring to 2021 confirms compression persisted beyond the immediate pandemic adjustment, but the design cannot decompose how much reflects AI-driven substitution versus prolonged pandemic normalisation. HILDA also covers formal employment only, leaving out gig and informal workers who may carry different AI exposure profiles.

What the design needs is exogenous variation in AI adoption that is independent of the pandemic cycle. Differential uptake of large language models across occupations after ChatGPT’s release in late 2022 is one candidate instrument (Acemoglu & Restrepo, 2018). Extending the HILDA panel through 2027 would shift identification weight toward a period where pandemic-era distortions are less likely to persist — and may be enough to separate the two mechanisms.

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